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Morphological and genetic diversity of maize landraces along an altitudinal gradient in the Southern Andes

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Abstract

Maize (*Zea mays* ssp. *mays)* is a major cereal crop worldwide and is traditionally or commercially cultivated almost all over the Americas. The North-Western Argentina (NWA) region constitutes one of the main diversity hotspots of the Southern Andes, with contrasting landscapes and a large number of landraces. Despite the extensive collections performed by the "Banco Activo de Germoplasma INTA Pergamino, Argentina" (BAP), most of them have not been characterized yet. Here we report the morphological and molecular evaluation of 30 accessions collected from NWA, along an altitudinal gradient between 1120 and 2950 meters above sea level (masl). Assessment of morphological variation in a common garden allowed the discrimination of two groups, which differed mainly in endosperm type and overall plant size. Although the groups retrieved by the molecular analyses were not consistent with morphological clusters, they showed a clear pattern of altitudinal structuring. Affinities among accessions were not in accordance with racial assignments. Overall, our results revealed that there are two maize gene pools co-existing in NWA, probably resulting from various waves of maize introduction in pre-Columbian times as well as from the adoption of modern varieties by local farmers. In conclusion, the NWA maize landraces preserved at the BAP possess high morphological and molecular variability. Our results highlight their potential as a source of diversity for increasing the genetic basis of breeding programs and provide useful information to guide future sampling and conservation efforts.

Introduction

Genetic erosion is the main problem associated with the selection and improvement of agronomically important species. Despite the progress achieved in terms of productivity, modern

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breeding takes advantage of only a small fraction of the available variability, restricting the response of crops to pests, diseases and environmental changes [1–3]. The use of landraces or related wild species provides the opportunity to counteract this process, widening the narrow genetic base of elite germplasm.

The recent recognition of landraces as sources of diversity and beneficial alleles has been expressed through different initiatives aimed at providing an extensive characterization of germplasm bank collections. This characterization involves molecular and phenotypic aspects of maize and other crops [e.g., https://seedsofdiscovery.org/; www.amaizing.fr; 4].

The high genetic diversity of maize landraces in the Americas has been thoroughly documented. [e.g. 5–12]. However, most of these studies have overlooked fine-scale variability and relied upon a limited number of accessions, and individuals, to represent large geographic areas. In addition, in contrast to the wealth of information regarding molecular variation, phenotypic characterization of landraces at their native sites is rarely reported. Both these facts reflect the still fragmentary nature of the knowledge needed for the efficient conservation and utilization of maize landraces.

The northwestern region of Argentina (NWA) is the southernmost distribution limit of the Andean maize landraces. This region is characterized by a remarkable topographic variability, as it comprises six phytogeographic provinces located within a relatively limited area (i.e, Yungas, Chaco, Puna, Pre-Puna, Monte and High Andean) [13]. Two contrasting examples of its great environmental diversity are the subtropical forests of the Yungas (distributed between 400 and 3000 masl and with annual mean precipitation increasing between 600 and 3000 mm with altitude), and the ridges and valleys of the Sub-Andean mountains (precipitation below 200 mm/ year, mainly concentrated in the summer months) [13].

More than 50% of the *circa* 56 maize landraces described for northern Argentina are native to NWA, making it one of the main diversity hotspots of the Southern Andes [14, 15]. In accordance, the most ancient records of maize from the American Southern Cone correspond to NWA and date to 3500 years before present (BP) [16]. Currently, native germplasm is traditionally cultivated between 400 and 3600 masl, thus being exposed to a wide range of thermal amplitudes and rainfall regimes [14]. So far, only a limited number of maize landraces from NWA have been characterized either molecularly and/or cytogenetically [17, 18]. The affiliations inferred from these studies showed a clear association of NWA landraces with the Andean group, originally defined by Mc. Clintock et al. [19] on the basis of the distribution of heterochromatic knobs, and subsequently retrieved by Matsuoka et al. [12] and Vigouroux et al. [5] using microsatellite markers. However, several NWA accessions also showed evidence of introgression with germplasm from different racial or geographic origins (i.e. popcorns and tropical lowland maize), attesting to the heterogeneity of the gene pools currently found in this region [17, 18].

The present scenario of increasing food demand and climate change highlights the need for materials capable of growing under extreme conditions, such as the NWA landraces. Therefore, the characterization of the accessions in germplasm banks is important in tackling these challenges. Today, the "Banco Activo de Germoplasma de maíz in EEA-INTA Pergamino, Argentina" (BAP), preserves more than 2500 accessions from over 10 collections performed between 1977 and 1994. The almost complete absence of molecular information and the poor knowledge of the morphological variation and genetic merit of this vast collection [2, 20] hinders its utilization in breeding programs and neglects critical data to inform germplasm protection policies.

In this study, we present the morphological and molecular characterization of 30 accessions of BAP from NWA, which are representative of 17 NWA landraces and were collected at sites between 1120 and 2950 masl. They were sown in a common garden at 2300 masl and

characterized for 19 morphological characters and 22 SSR loci to evaluate their agronomic potential, determine their genetic constitution and guide *in situ* and *ex situ* conservation efforts.

Materials and methods

Plant material

A set of 30 accessions from BAP were selected to encompass the morphological diversity occurring in NWA and cover a broad altitudinal range (collection sites between 1120 and 2950 masl) (Fig 1, S1 Table). All the accessions were included in the morphological and molecular

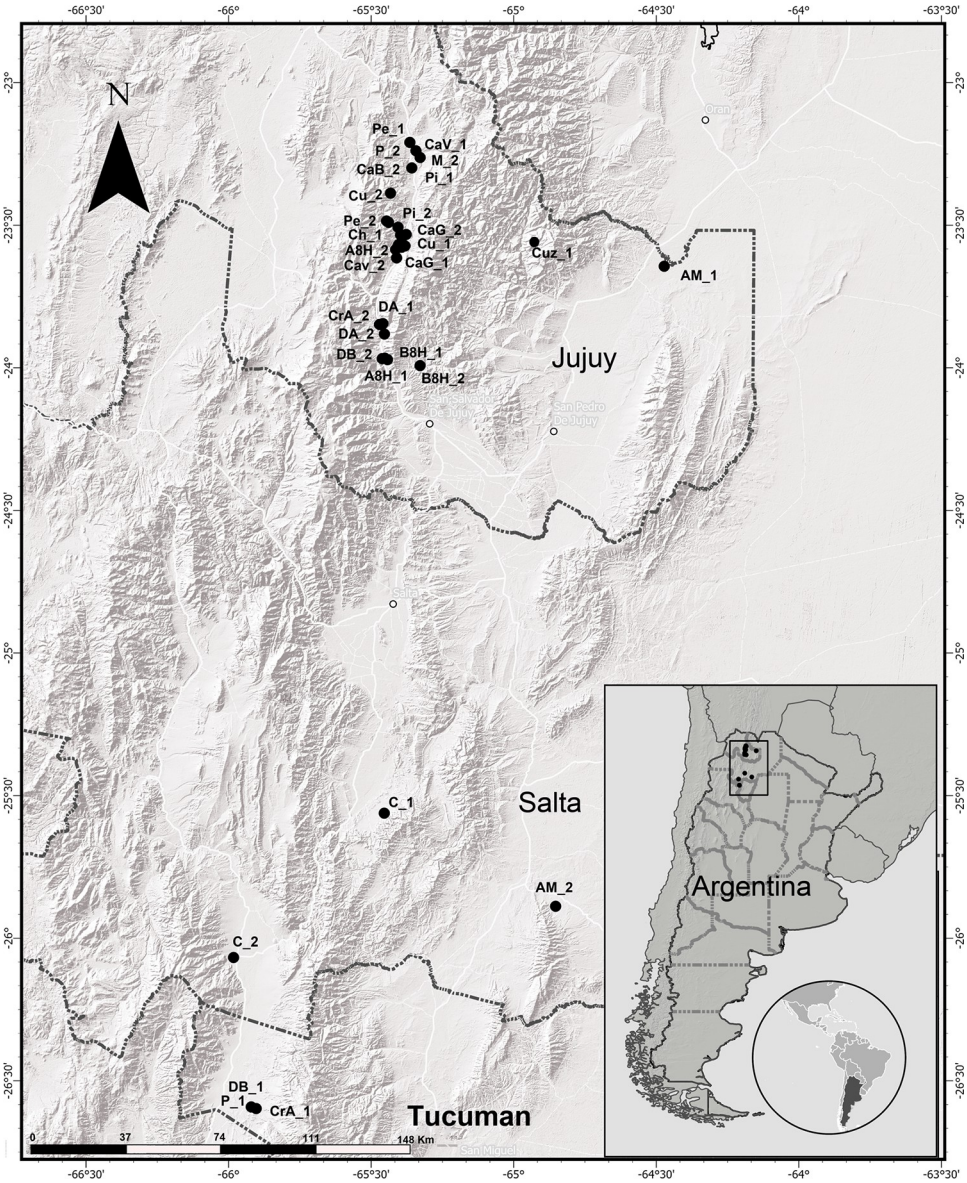


Fig 1. Collection sites of the landraces included in this study. Further details are provided in S1 Table. Map images were obtained from Aster Global Digital Elevation Model V003 (https://catalog.data.gov/dataset/aster-global-digital-elevation-model-v003).

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analyses. Throughout this study, the term landrace was used to define a taxonomic entity delimited by a set of morphological attributes according to the classification of Cámara Hernández et al. [14].

Morphological characterization

All the accessions were morphologically characterized at the Instituto de Pequeña Agricultura Familiar (IPAF), Hornillos, Jujuy Province, Argentina (23˚65'17" S, 65˚43'55"W; 2300 masl). One-hundred seeds per accession were sown under a randomized block design with two replicates. The elementary plots consisted of two rows of 5 m length, spaced 0.5 m apart. A conventional tillage was applied, with manual weeding and supplemental irrigation. No fertilizer or insecticide was added.

Nineteen agro-morphological descriptors, selected from the list of descriptors of the International Board for Plant Genetic Resources (CIMMYT/IBPGR, Roma, 1991, https://www.bioversityinternational.org/e-library/publications/detail/descriptors-for-maizedescriptores-para-maizdescripteurs-pour-le-mais/), were scored. The following vegetative traits were assessed: plant height (PH), number of leaves (NL), number of leaves above the uppermost ear (NLA), uppermost leaf length (ULL), uppermost leaf width (ULW), venation index (VI) and tillering index (TI). The descriptors measured on the tassel were tassel length (TL), tassel peduncle length (TPL), tassel branching space (TBS), number of primary branches on tassel (NPBT), number of secondary branches on tassel (NSBT) and number of tertiary branches on tassel (NTBT). The variables recorded on the ear were uppermost ear height (EH), ear peduncle length (EPL), ear diameter (ED), mean number of ears per accession (MNE), number of rows of kernels per ear (NRK), and number of kernels per row (NKR).

Vegetative and tassel descriptors were measured on seven to ten plants per accession per block. The ear and kernel descriptors were measured on ten ears per accession, which were harvested from the same plants used to evaluate vegetative and tassel descriptors.

Morphological data analysis

Summary statistics, coefficients of variation (CV) and correlations between traits were computed using InfoStat 2018 [21].

Using the accessions as operational taxonomic units (OTUs), a Principal Component Analysis (PCA) was performed based on average trait values. To avoid biases related to the difference in scale between the variables, data were standardized so that their average was zero and the standard deviation was equal to one. R packages *FactoMineR* [22] and *factoextra* [23] were used to compute and visualize PCA results. Groups of accessions were identified by the K-means algorithm [24], employing the Bayesian Information Criterion (BIC) [25] to find the most likely number of clusters. These analyses were conducted with the R package *adegenet* [26].

Microsatellite genotyping

A total of 379 individuals were genotyped for the molecular analyses. The number of individuals per accession is given in S1 Table. Genomic DNA was extracted from lyophilized young leaves (2–3 days old), germinated from individual kernels following Dellaporta et al. [27]. The quality and concentration of the genomic DNA was assessed using NanodropND1000 3.3 software (NanoDrop Technologies®).

Twenty-two SSR loci were selected from a preliminary survey of 27, and only loci with unambiguous interpretation were used for this analysis (S2 Table). Genotyping of the SSRs was performed using PCR with fluorescent labeled primers (HEX and FAM). PCR products

were size-separated on an Applied Biosystems automated sequencer (ABI 3130 XL) and allele calling was carried out with GeneMapper® 4.0 software (Applied Biosystems, Foster City, USA) using a commercial size standard for allele size assignment (GeneScan ROX 500, Applied Biosystems®). Automatic allele calls were subsequently confirmed reviewing all electropherograms.

Microsatellite data analysis

Mean number of alleles per locus (A), allele frequency, observed (H_o) and expected (H_e) heterozygosities, allelic richness (R_s) [28], presence of population-specific alleles (hereafter referred to as private alleles) and Wright's fixation indices were assessed using Fstat 2.9.3.2 software [29]. Genetic distances among accessions were computed according to Nei [30] using GeneAlEx 6 (Peakall & Smouse, 2006).

Genetic structure was examined using the Bayesian model-based approach of Pritchard et al. [31] implemented in STRUCTURE 2.3.4 software (http://www.pritch.bsd.uchicago.edu). The number of clusters evaluated ranged from 1 to 10. The analysis was performed using 10 replicates runs per K value, a burn-in period length of 500,000 and a run length of 1,000,000. No prior information on the origin of individuals was used to define the clusters and the correlated frequency model was used for all the analyses. The model assumes that the frequencies in the different populations are likely to be similar due to common ancestry [32]. The deltaK method of Evanno et al. [33] was used to identify the most likely number of clusters, using the web tool STRUCTURE HARVESTER [34]. Accessions were assigned to a given cluster when average membership coefficients were higher than an arbitrary cut-off value of 85%. Graphical display of STRUCTURE outputs was performed via the distruct program version 1.1 [35].

Correlations between genetic, morphological, and altitudinal distances were evaluated using Mantel tests (Mantel, 1964). Analyses were conducted based on Spearman correlation coefficient using the *mantel* function of the R package *vegan* [36]. We also inspected Spearman correlation between altitude, morphological traits and genetic diversity measures using InfoStat version 2018 [21]. Significance was assessed with permutation tests (1000 permutations). All graphics were obtained with the R package *ggplot2* [37].

Results

Morphological variation

Global averages for the 19 quantitative agro-morphological traits scored in the study are presented in Table 1, along with standard deviations and coefficients of variation. Data for individual accessions are provided in S3 Table.

All the morphological characters analyzed showed considerable levels of variation across accessions (Table 1). The coefficient of variation was highest for TI (73%), NSBT (46%) and TPL (30%), while the most homogeneous traits were NLA, ULW and ED, with coefficients of variation of 9, 8 and 8%, respectively.

The analysis of association between variables revealed 56 significant correlations, which decreased to 20 after Bonferroni correction (S4 Table). Most of the significant correlations were moderate, except for PH-EH (r = 0.87) and TBS- NSBT (r = 0.76).

Analysis of morphological variation using the K-means algorithm allowed the identification of two main groups (K = 2, BIC = 86.43, diffNgroup criterion) (S3 Table). Accession assignment and distribution along the first two dimensions of the PCA are shown in Fig 2A. The first PC accounted for 32.6% of the variance and was negatively associated with the variables that made the largest contributions to this component, i.e. PH, EH, TBS, and NLA (Fig 2B).

Table 1. Quantitative trait variation in 30 maize accessions from Northwestern Argentina (NWA).

Trait	Mean	Standard deviation	Coefficient of variation
PH (cm)	164.16	22.94	14
NL	12.38	1.63	13
NLA	5.99	0.53	9
ULL (cm)	65.68	7.69	12
ULW (cm)	9.26	0.78	8
VI	2.50	0.35	14
TI	0.86	0.63	73
TL (cm)	35.41	7.51	21
TPL (cm)	17.48	5.21	30
TBS (cm)	13.15	3.31	25
NPBT	16.38	3.10	19
NSBT	6.64	3.05	46
NTBT	0.23	0.38	16.5
EH (cm)	81.55	18.93	23
EPL (cm)	8.30	2.13	26
ED (cm)	4.01	0.31	8
MNE	2.55	0.43	17
NRK	12.26	2.08	17
NKR	30.53	4.13	13

PH: plant height, **NL**: number of leaves, **NLA:** number of leaves above the uppermost ear, **ULL**: uppermost leaf length, **ULW**: uppermost leaf width, **VI**: venation index, **TI**: tillering index, **TL**: tassel length, **TPL**: tassel peduncle length, **TBS**: tassel branching space, **NPBT**: number of primary branches on tassel, **NSBT**: number of secondary branches on tassel, **NTBT**: number of tertiary branches on tassel, **EH**: uppermost ear height, **EPL**: ear peduncle length, **ED**: ear diameter, **MNE**: mean number of ears, **NRK**: number of rows of kernels per ear, **NKR**: number of kernels per row

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The second PC accounted for 13.9% of total variance. It was positively associated with NTBT and negatively correlated with TI, MNE, and ULL.

The first group, G1 (black), presented a heterogeneous grain conformation, with accessions showing dent, semi-dent, sweet, flint and floury endosperm types (S1 Table). In this cluster, plants were taller and leafier than those in the second group, G2 (blue). In addition, tassel peduncles were shorter, the tillering index was lower, ear diameter was larger and the average number of grains per ear was 372 (S3 Table).

On the other hand, the G2 group comprised low-height accessions with pop, flint or semi-flint endosperm, fewer leaves, longer tassel peduncles, shorter branching space and very few tassel secondary branches. These accessions produced only a few ears per plant. The ears had smaller mean diameter and fewer grains per row than those in group G1, but a larger number of rows of kernels, resulting in a mean number of grains per ear of 350 (S3 Table).

The third PC (12.1% of total variance) maintained the same groupings, but further divided G1 into two sub-clusters, separating accessions with higher TPL and lower VI from the rest (S1 Fig, S5 Table).

Molecular diversity

Genotyping of 22 SSR loci in 379 individuals revealed a total of 419 alleles, with sample sizes per accession ranging from 7 to 17 individuals (S1–S6 Tables). Diversity estimates per locus

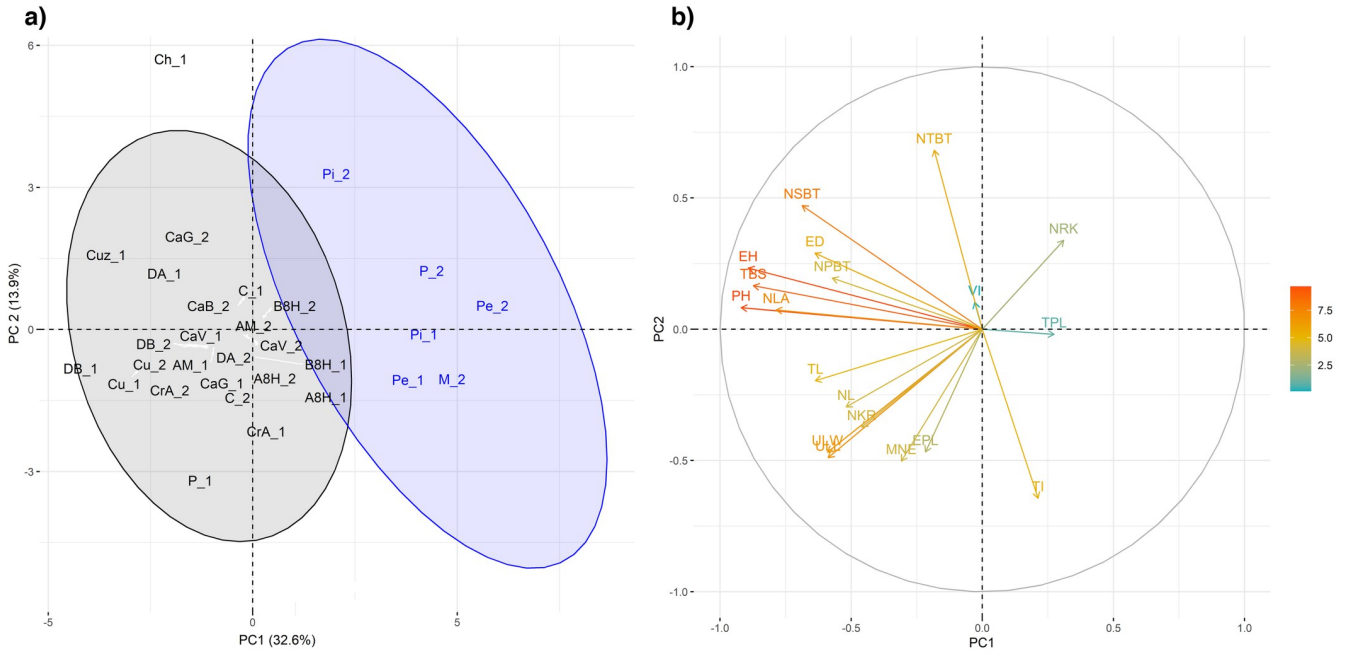


Fig 2. Principal Component Analysis based on 19 agro-morphological traits. (a) Scatter plot of PC1 vs. PC2. Accessions are color-coded according to the groups identified by the K-means procedure. (b) Correlations and variable contributions to the first two PCs.

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and accession are provided in S7 Table. The overall number of alleles per locus varied from eight (phi072) to 42 (bnlg244), with a mean of 19.05. The mean number of alleles per locus within accessions was 6.51, ranging from 4.23 (Pi_1) to 8.86 (B8H_2) (Table 2). The index H_o varied between 0.52 (CaG_1) and 0.65 (Pi_2), H_e between 0.63 (Pi_1) and 0.80 (B8H_2), and Rs between 3.77 (Pi_1) and 5.36 (B8H_2) (Table 2). A total of 35 private alleles were found distributed in 73.3% of the accessions (Table 2), with only four of them showing frequencies higher than 0.1. In regard to He, A and Rs, the most variable accession was B8H_2, while AM_1, Pe_2, DB_1 and Pi_2 exhibited the highest number of private alleles (Table 2). All accessions showed deviations from panmixia, with an excess of homozygotes (Table 2).

Population structure

STRUCTURE-model-based approach allowed discrimination of different genetic clusters within the group of accessions examined here (Fig 3). Using the method of Evanno et al. (2005), maximal ΔK occurred at K = 2, with the next largest peak at K = 3 (S8 Table).

For K = 2, most accessions could be assigned to one of the two groups, with an average proportion of membership higher than 0.85, except for A8H_1 and B8H_2, which showed an admixed composition (Fig 3A), consistent with the high variability indices observed for these accessions. The genetic groups did not match with those obtained with the morphological analysis. The first cluster, yellow, included 10 of the 24 accessions assigned to group G1, while the second cluster, dark purple, included all the accessions of the morphological group G2 and 12 of the group G1 (Fig 3A). When considering K = 3, the accessions belonging to the yellow cluster of K = 2 still formed a unit, while those of the dark purple cluster became separated into two clusters, blue and pink, one of which coincided with the morphological group G2 (Fig 3B).

A detailed examination of Fig 3A revealed that the genetic groups were associated with the altitude of the collection site, with a clear distinction between accessions cultivated below and

Table 2. Genetic variability in maize landraces from Northwestern Argentina (22 SSR loci).

		A		Ho		He		Rs		PA	F_{IS}
Accession	**N**	**M**	**s.d.**	**M**	**s.d.**	**M**	**s.d.**	**M**	**s.d.**		
AM_1	14	5.59	3.54	0.57	0.32	0.66	0.24	3.99	1.74	4	0.14**
AM_2	13	5.91	3.28	0.64	0.22	0.70	0.17	4.22	1.46	1	0.09*
C_1	8	5.27	2.62	0.59	0.20	0.72	0.17	4.39	1.79	0	0.18**
C_2	14	6.77	2.89	0.60	0.23	0.75	0.12	4.56	1.32	0	0.19**
P_1	13	8.14	3.54	0.64	0.18	0.79	0.12	5.29	1.55	2	0.20**
P_2	11	5.41	3.29	0.63	0.28	0.71	0.21	4.23	1.78	0	0.11**
Pe_1	7	4.82	2.24	0.52	0.27	0.67	0.22	4.22	1.73	0	0.23*
Pe_2	15	6.32	3.67	0.55	0.23	0.67	0.24	4.21	1.77	3	0.17**
DB_1	12	7.41	2.79	0.58	0.18	0.79	0.17	5.21	1.43	3	0.27**
DB_2	14	7.27	3.28	0.62	0.18	0.77	0.12	4.90	1.54	1	0.20**
A8H_1	14	7.45	3.25	0.63	0.19	0.76	0.15	4.90	1.50	1	0.18**
A8H_2	15	7.55	4.42	0.62	0.22	0.72	0.23	4.77	1.96	1	0.13**
CaB_2	15	6.36	3.22	0.60	0.24	0.67	0.24	4.26	1.64	1	0.10**
DA_1	15	7.36	3.06	0.61	0.24	0.77	0.13	4.92	1.39	2	0.21**
DA_2	15	7.86	4.60	0.59	0.23	0.70	0.21	4.66	1.91	1	0.16**
CaV_1	12	6.50	3.69	0.55	0.25	0.69	0.25	4.54	1.91	0	0.21**
CaV_2	14	6.77	3.58	0.59	0.23	0.70	0.24	4.60	1.86	2	0.15**
CrA_1	7	5.52	2.02	0.54	0.19	0.79	0.15	4.88	1.50	1	0.31**
CrA_2	15	7.23	4.39	0.63	0.23	0.69	0.24	4.53	1.88	1	0.09**
B8H_1	14	7.00	3.19	0.61	0.20	0.75	0.13	4.69	1.42	0	0.18**
B8H_2	17	**8.86**	3.67	0.62	0.18	**0.80**	0.14	**5.36**	1.47	1	0.23**
M_2	13	5.86	3.08	0.58	0.26	0.66	0.25	4.11	1.70	0	0.10**
CaG_1	8	5.59	2.97	0.52	0.25	0.70	0.28	4.64	2.01	1	0.27**
CaG_2	16	6.86	3.51	0.58	0.26	0.67	0.25	4.32	1.72	1	0.13**
Pi_1	7	4.23	2.49	0.53	0.30	0.63	0.24	3.77	1.97	0	0.12*
Pi_2	14	6.36	3.57	**0.65**	0.23	0.72	0.20	4.45	1.61	3	0.10**
Cuz_1	11	5.95	3.53	0.53	0.27	0.65	0.27	4.33	2.06	1	0.18**
Cu_1	10	5.82	3.23	0.58	0.19	0.69	0.23	4.41	1.87	2	0.16**
Cu_2	15	6.86	3.60	0.62	0.23	0.70	0.20	4.44	1.64	1	0.11**
Ch_1	11	6.38	3.35	0.57	0.27	0.71	0.22	4.55	1.76	1	0.19**

N: number of individuals. A: number of alleles, Ho: observed heterozygosity, He: expected heterozygosity, Rs: mean allelic richness. PA: private alleles. M: mean, s.d.: standard deviation. The highest value of each index is highlighted in bold. * $p<0.05$; ** $p<0.001$.

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above 2000 masl (Fig 3A). Fig 4A shows the proportions of membership to the dark purple cluster (K = 2) in function of the altitude of the collection site. Although the correlation was significant ($r_{Spearman} = 0.7$, $p < 0.001$), the pattern was not gradual, exhibiting an abrupt jump above 2000 masl. When these data were overlapped with the assignments found for K = 3, the accessions belonging to the yellow group corresponded to the lowest part of the altitude gradient, while those of the pink and blue clusters were cultivated above 2000 masl. Indeed, for K = 3, the mean altitude of the collection site of the accessions within the yellow, pink and blue clusters were 1684, 2584 and 2745 masl, respectively (Fig 4B).

The neighbor-joining network based on Nei's distance was consistent with the partition into two groups inferred by STRUCTURE (S2 Fig). In most cases, at the level of individual accessions, no clustering was observed between accessions of a same landrace. The overall difference between accessions was moderate, with $F_{ST} = 0.092$ ($p < 0.01$).

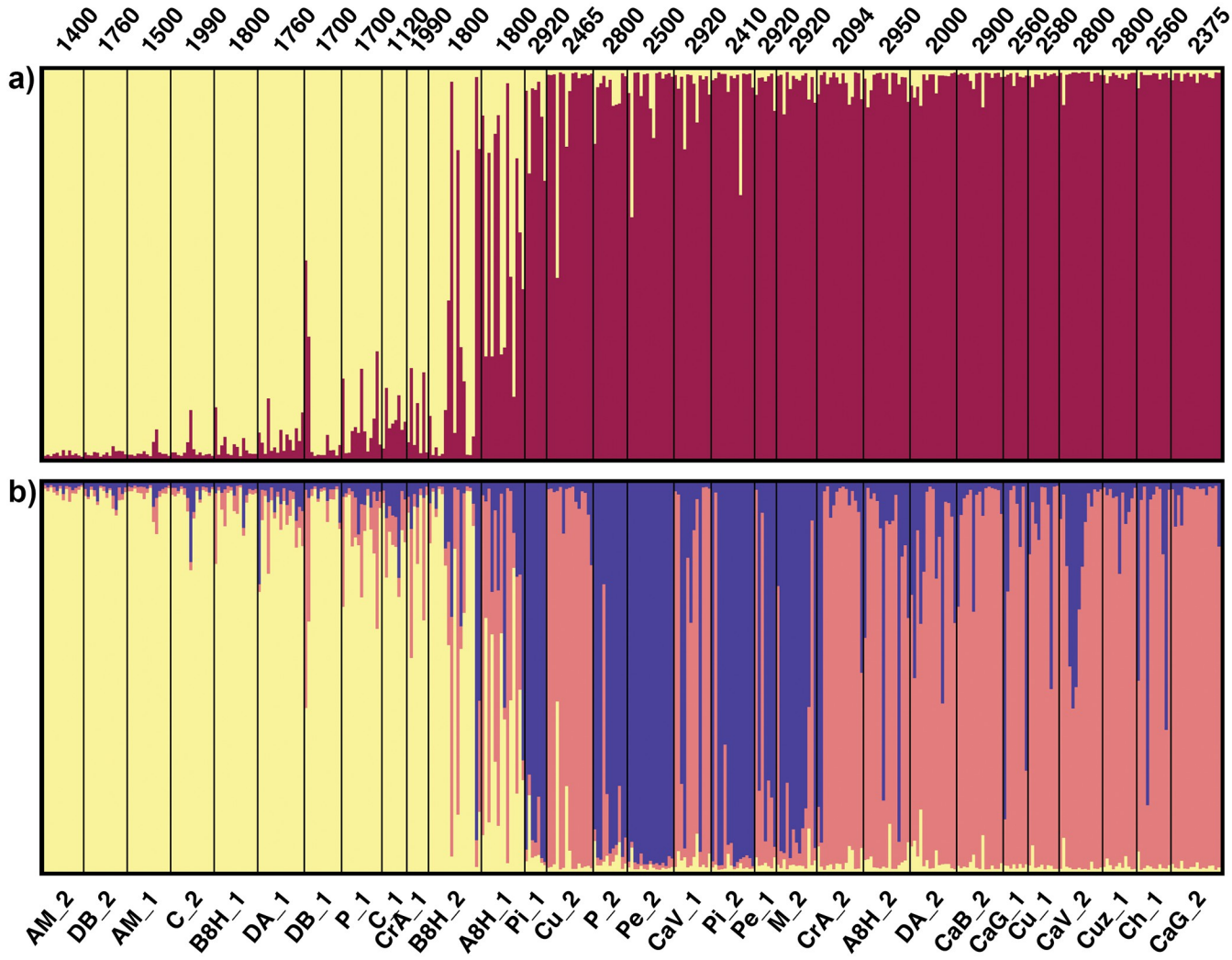


Fig 3. Population structure of maize accessions from Northwestern Argentina (NWA). (a) STRUCTURE bar plot for K = 2. (b) STRUCTURE bar plot for K = 3. Each vertical line represents an individual and colors represent their inferred ancestry from K ancestral populations. Accessions are ordered by altitude of the collection site.

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Isolation by distance and clinal variation

Morphological distances showed no correlation with genetic, geographic or altitudinal distances among accessions (Fig 5; Mantel test, $p>0.05$). Conversely, there were significant associations between genetic and altitudinal distances (Fig 5A), as well as between genetic and geographic distances (Fig 5B) (Mantel test, $r = 0.44$, $p<0.001$: $r = 0.54$, $p<0.001$ respectively).

Despite the absence of correlation between morphological distance and altitude, the characters NL and NKR were negatively associated with the latter (Table 3). Likewise, a decrease in genetic variability was observed with altitude for the indices He and Rs (Fig 6A and 6B, respectively), but not for the estimators A and PA (Spearman correlation, $p>0.05$).

Discussion

Efficient use and conservation of the landraces available in germplasm banks requires comprehensive genetic and agronomic characterization focused primarily on variation at the level of

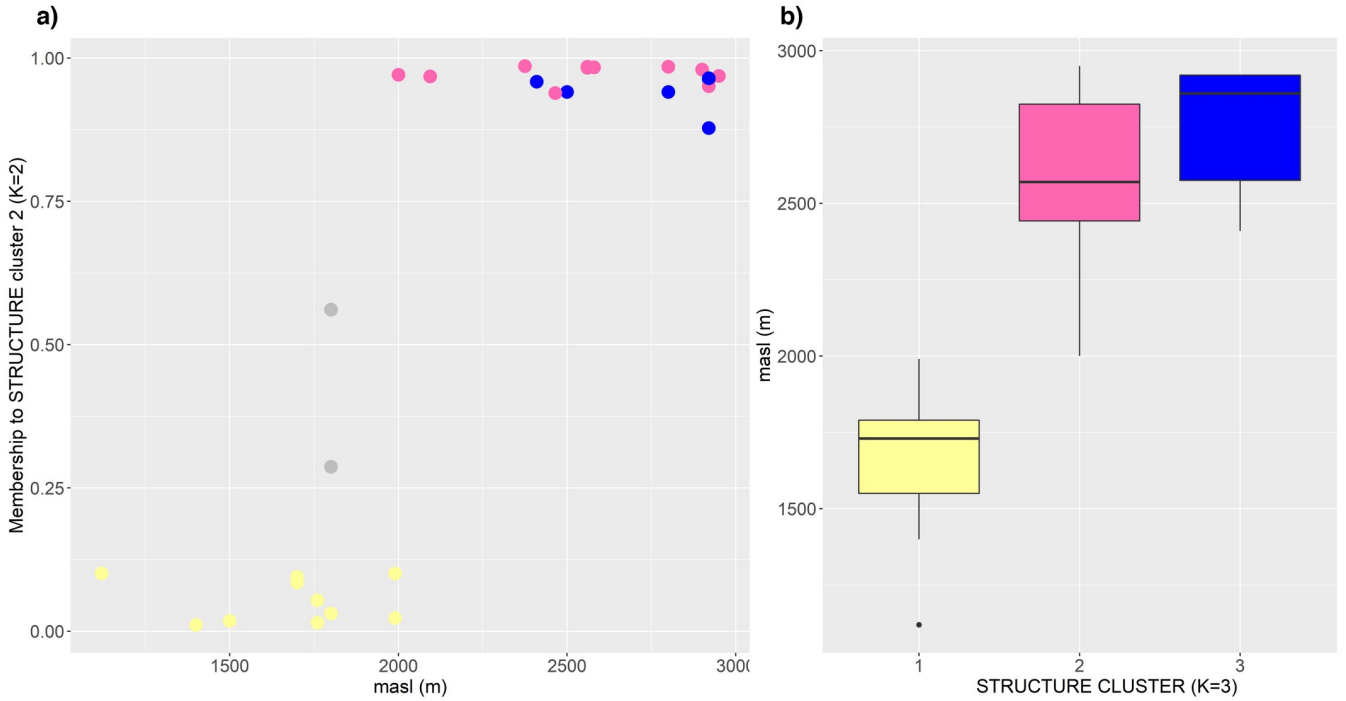


Fig 4. Relationship between population structure and altitude of the collection site. (a) Scatterplot of altitude vs. the inferred ancestry to STRUCTURE cluster 2 (K = 2). (b) Box plots of altitude of collection site for the groups inferred with K = 3.

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individual accessions [3, 38–40]. Consistent with this idea, our results suggest that racial assignment, even when performed according to the standardized criteria of germplasm banks, is a poor indicator of similarity between accessions, at both genetic and morphological levels. The lack of racial cohesion may be explained by the dynamics of production and exchange of landraces among local farmers. Small holders in NWA, just like in Mexico [41, 42], and most

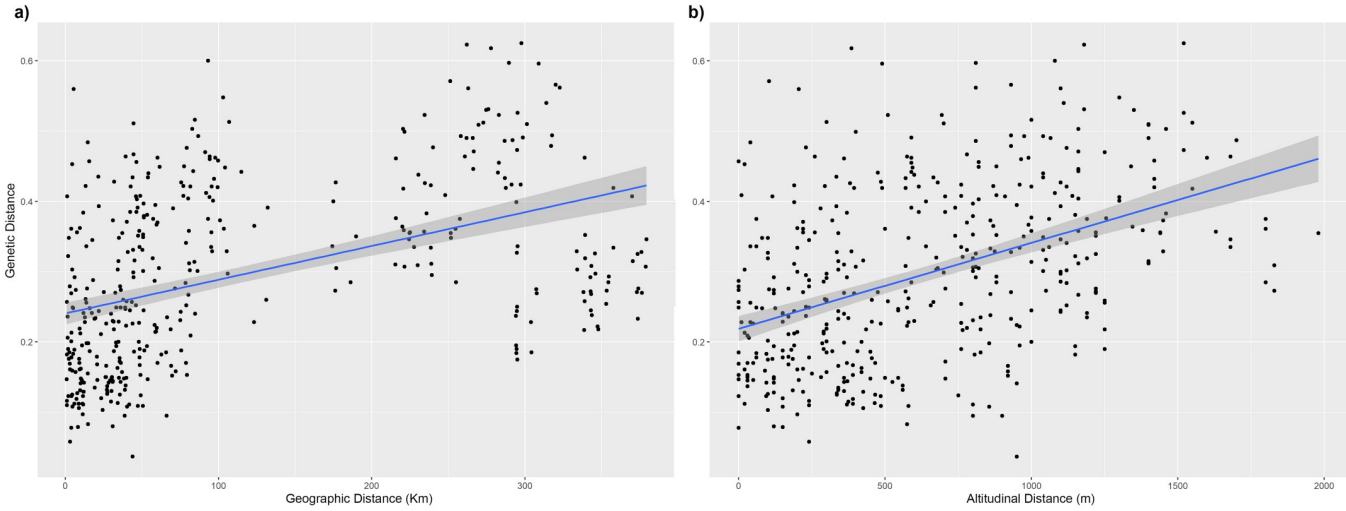


Fig 5. Mantel matrix correlation tests. (a) Scatterplot showing the correlation between genetic and geographic distances. (b) Scatterplot showing the correlation between genetic and altitudinal distances.

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Table 3. Spearman correlation between agro-morphological traits and altitude.

Trait	Spearman r	p-value
PH	-0.28	0.1394
EH	-0.32	0.0865
NLA	-0.37	0.0443
TI	-0.12	0.5269
NL	-0.61	**0.0004**
ULL	-0.09	0.62
ULW	-0.02	0.9017
VI	-0.41	0.0229
TL	-0.24	0.2014
TPL	0.39	0.0391
TBS	-0.26	0.1661
NPBT	-0.48	0.0087
NSBT	-0.11	0.5875
NTBT	0.4	0.027
EPL	-0.05	0.8084
MNE	1.20E-03	0.9948
NRK	0.06	0.7592
ED	-0.07	0.7072
NKR	-0.76	**<0.0001**

Correlation coefficients that remained significant after Bonferroni corrections are highlighted in bold (p <0.0026). PH: plant height, **NL**: number of leaves, **NLA:** number of leaves above the uppermost ear, **ULL**: uppermost leaf length, **ULW**: uppermost leaf width, **VI**: venation index, **TI**: tillering index, **TL**: tassel length, **TPL**: tassel peduncle length, **TBS**: tassel branching space, **NPBT**: number of primary branches on tassel, **NSBT**: number of secondary branches on tassel, **NTBT**: number of tertiary branches on tassel, **EH**: uppermost ear height, **EPL**: ear peduncle length, **ED**: ear diameter, **MNE**: mean number of ears, **NRK**: number of rows of kernels per ear, NKR: number of kernels per row.

https://doi.org/10.1371/journal.pone.0271424.t003

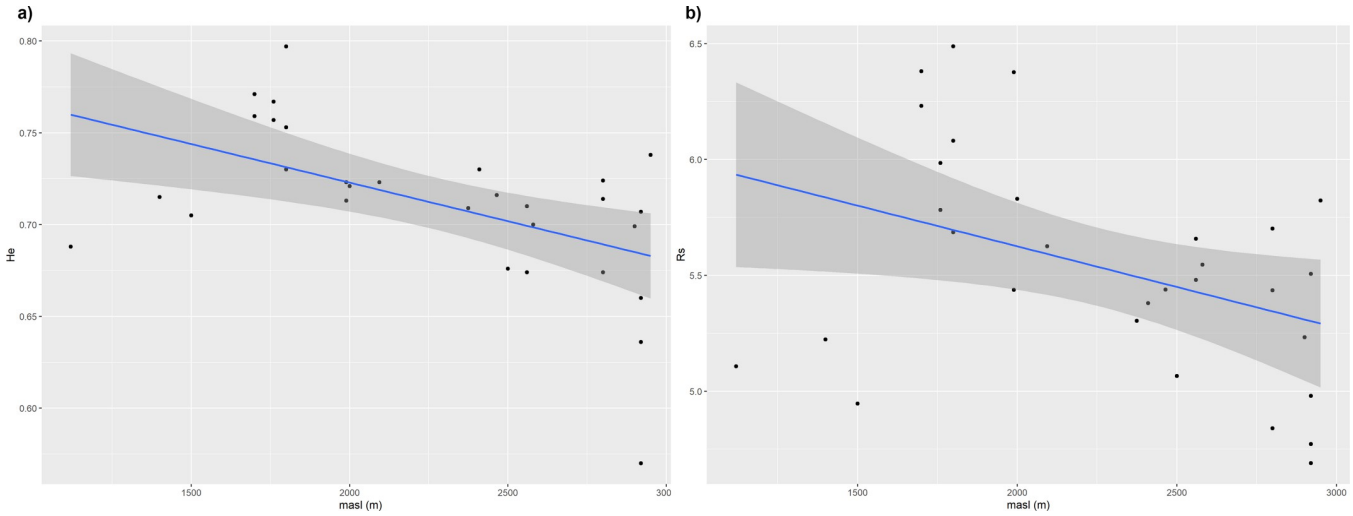


Fig 6. Correlation analysis between diversity estimates and altitude of the collection site. (a) Scatterplot of the correlation between altitude and expected Heterozygosity, He. (b) Scatterplot of the correlation between altitude and allelic richness, Rs.

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probably in the rest of Latin America, have adopted cultivation practices that allow unintentional racial crossing, as they grow different maize forms or varieties in small areas close to each other, thus facilitating pollen exchange [43]. At the same time, producers usually exchange grains, promoting gene flow through the dispersal of different maize forms among geographically distant localities. Moreover, the introduction of commercial or improved germplasm, which generally provides economic benefits for farmers, has also contributed to blur racial boundaries [14, 43]. In fact, the mixing of such germplasm with native landraces produces creolized varieties that farmers would later identify as "local" [14, 43].

In this study, the clusters derived from morphological characters are consistent with those previously described for NWA landraces. Melchiorre et al. [44, 45] identified two major maize groups: the first one, regarded as more primitive, is related to the popcorn type (Pisingallo) with early maturity, rather corneous grains and smaller plants; the second group is more heterogeneous, with totally or partially corneous or floury grains and larger plants. The landraces studied here display the same pattern, where groups are associated with the type of endosperm and the main discriminating characters are related to plant size (PH, EH, TBS and NLA).

The two major groups identified in the molecular analysis did not coincide with the morphological groups and showed a clear association with the altitude of collection sites, with the morphological group G2 being completely included in the highland group (>2000 masl). The levels of variability also appeared to go along with the differentiation of genetic groups, with a decrease in parameters such as expected heterozygosity and allelic richness in function of altitude. The lower variability of the highland group is consistent with the difficulties in growing maize under harsh environmental conditions [14, 43]. In the highlands, frequent frosts and low temperatures shorten flowering and maturation times, thus reducing seed yields [14, 43]. Consequently, population sizes tend to be smaller, intensifying the effect of genetic drift and promoting the loss of diversity within farmers' fields [43].

Regardless of how variability is distributed, NWA landraces taken as a whole show high levels of genetic variability, which are similar to the values recorded at the centers of crop diversity [5, 9, 10, 41, 46, 47]. For instance, when limiting the comparison to the subset of loci shared between our study and that of Vigouroux et al. [5], the mean number of alleles detected here (A = 19.05) is only slightly lower than the estimates reported by these authors for both the Highland Mexico (A = 20.4) and Andean (A = 21.45) groups. Likewise, the estimates of He for NWA accessions varied within a range previously reported for landraces from Mexico and other regions of the Americas (He: 0.61–0.81) [5, 9, 10, 41, 46, 47]. These results emphasize the potential value of the NWA landraces conserved in the BAP, since only 30 accessions contained almost the same number of alleles as germplasm coming from the central regions of maize distribution.

Several studies have found a relationship between altitude of collection site and genetic composition of maize landraces both at the regional and continental levels [8, 10, 48]. The existence of correlation between genetic differentiation and altitude may be interpreted as resulting from clinal variation, where the genetic distances increase gradually with increasing altitudinal distance. However, in the landraces analyzed here, the relationship between altitude and inferred ancestry to the clusters obtained by STRUCTURE (Fig 4A) suggests a more abrupt cutoff, where the accessions of intermediate position are those showing a signature of recent introgression.

On the other hand, although morphological differentiation was not found to be correlated with altitude, individual trait analysis revealed a significant decrease for NL and NRK. Considering that all measurements were made in a common garden experiment at 2300 masl, these results may be interpreted as a local adaptation rather than phenotypic plasticity in response to environmental conditions. In fact, the relationship between flowering time, which is a key

factor in the adaptation of landraces to altitude [49, 50], and NL has been extensively documented [51–55]. In particular, Li et al [56] showed that flowering time shared genetic determinants with the number of leaves below, but not above, the uppermost ear (NLA). These results may help explain why we found an association between NL, but not NLA, and altitude. As observed here for NL, the reduction in genome size with altitude has been proposed to be an indirect consequence of selection on flowering time [57]. In agreement with these hypotheses, NWA landraces also exhibit patterns of clinal variation for both the DNA content of autosomes and the occurrence of B chromosomes [18, 58]. Indeed, evidence of the adaptive significance of these clines was provided by Lia et al. [59].

One of the main findings of our study is the remarkable differentiation between lowland (<2000 masl) and highland (>2000 masl) NWA accessions (Fig 3). In agreement with this distinction, South American maize has traditionally been divided into Tropical Lowland and Andean landraces [5]. In turn, these two groups have been related to distinct events of maize dispersal into the continent, namely, landraces that entered through the eastern lowlands of the continent and those introduced from the Central Andes [12, 60]. More recently, Bracco et al. [9] and Kistler et al. [61] proposed the existence of three major maize lineages in South America. However, in contrast to the classical view [12, 60], Kistler et al. [61] suggested that two of these, the Andean and Lowland lineages, were derived from a partially domesticated maize that had been introduced to south-western Amazon at early stages, whereas the third lineage, the Pan-American, was introduced later through lowland South America. Although highland NWA landraces have been previously affiliated with the Andean group [9], further evidence is still needed to establish the relationship between the NWA genetic groups identified here and the lineages postulated by previous authors. In addition to the uncertainties concerning the movements of maize during pre-Columbian times, the influence of the varieties and hybrids resulting from modern breeding is another factor to be considered when interpreting the genetic composition of maize from NWA. In this regard, Cámara Hernández et al. [14] provided numerous examples of creolized landraces, particularly for the lowlands. Likewise, evidence obtained with chloroplast genome sequencing support the occurrence of introgression between traditional NWA landraces and more modernly improved varieties [62].

In conclusion, the present study constitutes a valuable contribution to existing *ex situ* conservation programs and to design future collection and management strategies. In particular, our results indicate that altitudinal structuring is a key factor in decision-making. For example, Perales et al. [63] reported that the replacement rate of native landraces by modern varieties in Mexico was considerably higher in lowlands and mid-elevation sites than in highlands because there were few modern varieties capable of outperforming native materials in high-altitude environments. These findings led them to suggest that interventions should favor *in situ* conservation strategies in lowland or intermediate elevation sites, where modern varieties have largely outcompeted native landraces. Despite the lack of similar studies in NWA landraces, it is clear that the altitudinal structuring of genetic diversity constitutes a relevant factor to be considered for their conservation.

Supporting information

S1 Fig. Principal Component Analysis of morphological variation. PC1 vs. PC3.
(TIF)

S2 Fig. Neighbor-joining tree based on SSR data.
(PDF)

S1 Table. Details of the accessions used in this study.
(DOCX)

S2 Table. Primers used in the present study.
(DOCX)

S3 Table. Morphological data per accession.
(XLSX)

S4 Table. Correlation analyses of morphological traits.
(DOCX)

S5 Table. Correlation between morphological traits and the first three axes of the Principal Component Analysis.
(XLSX)

S6 Table. SSR data matrix.
(XLSX)

S7 Table. Genetic_diversity estimates per locus and accession.
(XLSX)

S8 Table. Evanno deltaK.
(DOCX)

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References

1. Hoisington D, Khairallah M, Reeves T, Ribaut JM, Skovmand B, Taba S, et al. Plant genetic resources: What can they contribute toward increased crop productivity? Proc Natl Acad Sci U S A. 1999; 96: 5937–5943. https://doi.org/10.1073/pnas.96.11.5937 PMID: 10339521
2. Eyherabide G, Lorea R, Delucchi C, Lopez C, Ferrer M, Presello D, et al. Evaluación del potencial de variedades nativas de maíz como recurso para el mejoramiento de un cruzamiento entre líneas duras coloradas. Actas VIII Congreso Nacional de Maíz. 2005; 1: 354–357.
3. McCouch S, Baute GJ, Bradeen J, Bramel P, Bretting PK, Buckler E, et al. Feeding the future. Nature. 2013; 499: 23–24.
4. Lopes MS, El-Basyoni I, Baenziger PS, Singh S, Royo C, Ozbek K, et al. Exploiting genetic diversity from landraces in wheat breeding for adaptation to climate change. J Exp Bot. 2015; 66: 3477–86. https://doi.org/10.1093/jxb/erv122 PMID: 25821073
5. Vigouroux Y, Glaubitz JC, Matsuoka Y, Goodman MM, Sánchez G. J, Doebley J. Population structure and genetic diversity of New World maize races assessed by DNA microsatellites. Am J Bot. 2008; 95: 1240–1253. https://doi.org/10.3732/ajb.0800097 PMID: 21632329
6. van Heerwaarden J, Doebley J, Briggs WH, Glaubitz JC, Goodman MM, de Jesus Sanchez Gonzalez J, et al. Genetic signals of origin, spread, and introgression in a large sample of maize landraces. Proc Natl Acad Sci U S A. 2011; 108: 1088–92. https://doi.org/10.1073/pnas.1013011108 PMID: 21189301
7. Mir C, Zerjal T, Combes V, Dumas F, Madur D, Bedoya C, et al. Out of America: tracing the genetic footprints of the global diffusion of maize. Theor Appl Genet. 2013; 126: 2671–82. https://doi.org/10.1007/s00122-013-2164-z PMID: 23921956
8. Takuno S, Ralph P, Swarts K, Elshire RJ, Glaubitz JC, Buckler ES, et al. Independent Molecular Basis of Convergent Highland Adaptation in Maize. Genetics. 2015; 200:1297–312. https://doi.org/10.1534/genetics.115.178327 PMID: 26078279
9. Bracco M, Cascales J, Hernandez JC, Poggio L, Gottlieb AM, Lia VV. Dissecting maize diversity in lowland South America: Genetic structure and geographic distribution models. BMC Plant Biol. 2016 Aug 26. https://doi.org/10.1186/s12870-016-0874-5 PMID: 27561710
10. Bedoya CA, Dreisigacker S, Hearne S, Franco J, Mir C, Prasanna BM, et al. Genetic diversity and population structure of native maize populations in Latin America and the Caribbean. PLoS One. 2017; 12: 1–21. https://doi.org/10.1371/journal.pone.0173488 PMID: 28403177
11. Wang L, Beissinger TM, Lorant A, Ross-Ibarra C, Ross-Ibarra J, Hufford MB. The interplay of demography and selection during maize domestication and expansion. Genome Biol. 2017; 18: 215. https://doi.org/10.1186/s13059-017-1346-4 PMID: 29132403
12. Matsuoka Y, Vigouroux Y, Goodman MM, Sanchez G J, Buckler E, Doebley J. A single domestication for maize shown by multilocus microsatellite genotyping. Proc Natl Acad Sci U S A. 2002; 99: 6080–6084. https://doi.org/10.1073/pnas.052125199 PMID: 11983901
13. Oyarzabal M, Clavijo J, Oakley L, Biganzoli F, Tognetti P, Barberis I, et al. Unidades de vegetación de la Argentina. Ecol Austral. 2018; 28: 040–063.
14. Cámara Hernández J, Miante Alzogaray AM, Bellón R, Galmarini AJ. Razas de maíz nativas de la Argentina. Buenos Aires, Argentina: Editorial Facultad de Agronomía, Universidad de Buenos Aires; 2012.
15. Melchiorre P, Bartoloni N, Camara Hernadez J. Relaciones fenotipícas y genéticas entre razas tardías de maíz (*Zea mays ssp. mays*) nativas de la provincia de Misiones (Argentina). Bol la Soc Argentina Bot. 2020; 55: 23–43.
16. Oliszewski N, Arreguez G, Carrizo J. Identificiacion macro y microscopica de granos de *Zea mays* (Poaceae). Darwiniana. 2019; 7: 5–15.
17. Lia V V., Poggio L, Confalonieri VA. Microsatellite variation in maize landraces from Northwestern Argentina: Genetic diversity, population structure and racial affiliations. Theor Appl Genet. 2009; 119: 1053–1067.
18. Fourastié MF, Gottlieb AM, Poggio L, González GE. Are cytological parameters of maize landraces (*Zea mays ssp. mays*) adapted along an altitudinal cline? J Plant Res. 2018; 131: 285–296.
19. McClintock B, Kato TA, Blumenschein A. Chromosome Constitution of the Races of Maize, Its Significance in the Interpretation of Relationships Between Races and Varieties in the Americas. Chapingo, Mexico: Editorial Colegio de Postgraduados; 1981.
20. López C, Eyhérabide G, Delucchi C, Percibaldi N, Castellarín J, Pedrol H, et al. Selección de poblaciones locales de maíz como fuente de alelos favorables para el mejoramiento del peso hectolítrico y la relación de molienda en un híbrido flint x dentado. Actas VIII Congreso Nacional de Maíz. 2005; 1: 358–363.

21. Di Rienzo JA, Casanoves F, Balzarini M, Gonzalez L, Tablada M, Robledo CW. Infostat version 2018. Cordoba, Argentina: Centro de Transferencia Infostat, FCA, Universidad Nacional de Cordoba; 2018.

22. Lê S, Josse J, Husson F. FactoMineR: an R package for multivariate analysis. J Stat Softw. 2008; 25: 1–18.

23. Kassambara A, Mundt F. factoextra: Extract Version, Visualize the Results of Multivariate Data Analyses. R package. 2020.

24. Liu N, Zhao H. A non-parametric approach to population structure inference using multilocus genotypes. Hum Genomics. 2006; 2: 353–364. https://doi.org/10.1186/1479-7364-2-6-353 PMID: 16848973

25. Fraley C, Raftery AE. How Many Clusters? Which Clustering Method? Answers Via Model-Based Cluster Analysis. Comput J. 1998; 41: 578–588.

26. Jombart T. Adegenet: A R package for the multivariate analysis of genetic markers. Bioinformatics. 2008; 24: 1403–1405. https://doi.org/10.1093/bioinformatics/btn129 PMID: 18397895

27. Dellaporta SL, Wood J, Hicks JB. A plant DNA minipreparation: Version II. Plant Mol Biol Report. 1983; 1: 19–21.

28. El Mousadik A, Petit R. High level of genetic differentiation for allelic richness among populations of the argan tree endemic to Morocco. Theor Appl Genet. 1996; 92: 832–839.

29. Goudet J. FSTAT: a computer program to calculate F-Statistics. J Hered. 2013; 104: 586–590.

30. Nei M. Molecular Evolutionary Genetics. Arizona USA: Tempe AZ Arizona State University; 1987.

31. Pritchard JK, Stephens M, Donnelly P. Inference of population structure using multilocus genotype data. Genetics. 2000; 155: 945–959. https://doi.org/10.1093/genetics/155.2.945 PMID: 10835412

32. Falush D, Stephens M, Pritchard JK. Inference of population structure using multilocus genotype data: linked loci and correlated allele frequencies. Genetics. 2003; 164: 1567–87. https://doi.org/10.1093/genetics/164.4.1567 PMID: 12930761

33. Evanno G, Regnaut S, Goudet J. Detecting the number of clusters of individuals using the software STRUCTURE: a simulation study. Mol Ecol. 2005; 14: 2611–20. https://doi.org/10.1111/j.1365-294X.2005.02553.x PMID: 15969739

34. Earl D A, vonHoldt BM. STRUCTURE HARVESTER: A website and program for visualizing STRUCTURE output and implementing the Evanno method. Conserv Genet Resour. 2012; 4: 359–361.

35. Rosenberg N a. DISTRUCT: A program for the graphical display of population structure. Mol Ecol Notes. 2004; 4: 137–138.

36. Oksanen J, Guillaume Blanchet F. Friendly M, Kindt R, Legendre P, McGlinn D, Minchin PR, et al. vegan: Community Ecology Package. R package version 2.5–7 2020. https://CRAN.R-project.org/package=vegan.

37. Wickham H. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag New York; 2016.

38. Fowler C, Hodgkin T. Plant Genetic Resources for Food and Agriculture: Assessing Global Availability 1. Annu Rev Environ Resour. 2004; 29: 143–179.

39. Qualset CO, Shands HL. Safeguarding the future of US agriculture: The need to conserve threatened collections of crop diversity worldwide. University of California Genetic Resources Conservation Program. Div Agric Nat Resour Genet Resour Conserv Program, Davis, CA, USA; 2005.

40. Khoury C, Laliberté B, Guarino L. Trends in ex situ conservation of plant genetic resources: a review of global crop and regional conservation strategies. Genet Resour Crop Evol. 2010; 57: 625–639.

41. Pressoir G, Berthaud J. Patterns of population structure in maize landraces from the Central Valleys of Oaxaca in Mexico. Hered. 2003; 92: 88–94.

42. McLean-Rodríguez FD, Camacho-Villa TC, Almekinders CJM, Pè ME, Dell'Acqua M, Costich DE. The abandonment of maize landraces over the last 50 years in Morelos, Mexico: a tracing study using a multi-level perspective. Agric Human Values. 2019; 36: 651–668.

43. Ferrer M. Los recursos genéticos de maíz. In: Eyherabide G, editor. Bases para el manejo del cultivo del maíz. Buenos Aires, Argentina: Ediciones INTA; 2012. pp. 107–124.

44. Melchiorre P, Bartoloni N. Phenetic relationships among native races of maize (Zea mays spp. mays) from Northwestern Argentine (Catamarca). J Genet Breed. 2004; 58:233–242.

45. Melchiorre P, Bartoloni N, Camara Hernadez J. Relaciones fenéticas entre razas de maíz (Zea mays ssp. mays) nativas de la provincia de Jujuy (Argentina). Boletín la Soc Argentina Botánica. 2017; 52: 717–735.

46. Reif JC, Warburton ML, Xia XC, Hoisington DA, Crossa J, Taba S, et al. Grouping of accessions of Mexican races of maize revisited with SSR markers. Theor Appl Genet. 2006; 113: 177–185. https://doi.org/10.1007/s00122-006-0283-5 PMID: 16791685

47. Pineda-Hidalgo K V, Méndez-Marroquín KP, Alvarez EV, Chávez-Ontiveros J, Sánchez-Peña P, Garzón-Tiznado JA, et al. Microsatellite-based genetic diversity among accessions of maize landraces from Sinaloa in México. Hereditas. 2013; 150: 53–59.

48. Arteaga MC, Moreno-Letelier A, Mastretta-Yanes A, Vázquez-Lobo A, Breña-Ochoa A, Moreno-Estrada A, et al. Genomic variation in recently collected maize landraces from Mexico. Genomics Data. 2016; 7: 38. https://doi.org/10.1016/j.gdata.2015.11.002 PMID: 26981357

49. Romero Navarro JA, Willcox M, Burgueño J, Romay C, Swarts K, Trachsel S, et al. A study of allelic diversity underlying flowering-time adaptation in maize landraces. Nat Genet. 2017; 49: 476–480. https://doi.org/10.1038/ng.3784 PMID: 28166212

50. Wang L, Josephs EB, Lee KM, Roberts LM, Rellán-Álvarez R, Ross-Ibarra J, et al. Molecular Parallelism Underlies Convergent Highland Adaptation of Maize Landraces. Mol Biol Evol. 2021; 38: 3567–3580. https://doi.org/10.1093/molbev/msab119 PMID: 33905497

51. Colasanti J, Muszynski M. The Maize Floral Transition. In: Bennetzen JL, Hake SC, editors. Handbook of Maize: Its Biology. New York, USA: Springer; 2009. pp. 41–55.

52. Meng X, Muszynski MG, Danilevskaya ON. The FT-like ZCN8 gene functions as a floral activator and is involved in photoperiod sensitivity in maize. Plant Cell. 2011; 23: 942–960. https://doi.org/10.1105/tpc.110.081406 PMID: 21441432

53. Salvi S, Tuberosa R, Chiapparino E, Maccaferri M, Veillet S, Van Beuningen L, et al. Toward positional cloning of Vgt1, a QTL controlling the transition from the vegetative to the reproductive phase in maize. Plant Mol Biol. 2002; 48: 601–613. https://doi.org/10.1023/a:1014838024509 PMID: 11999837

54. Yang Q, Li Z, Li W, Ku L, Wang C, Ye J, et al. CACTA-like transposable element in ZmCCT attenuated photoperiod sensitivity and accelerated the postdomestication spread of maize. Proc Natl Acad Sci U S A. 2013; 110: 16969–16974. https://doi.org/10.1073/pnas.1310949110 PMID: 24089449

55. Li Z, Li K, Yang X, Hao H, Jing HC. Combined QTL mapping and association study reveals candidate genes for leaf number and flowering time in maize. Theor Appl Genet. 2021; 134: 3459–3472. https://doi.org/10.1007/s00122-021-03907-x PMID: 34247253

56. Li D, Wang X, Zhang X, Chen Q, Xu G, Xu D, et al. The genetic architecture of leaf number and its genetic relationship to flowering time in maize. New Phytol. 2016; 210: 256–268. https://doi.org/10.1111/nph.13765 PMID: 26593156

57. Bilinski P, Albert PS, Berg JJ, Birchler JA, Grote MN, Lorant A, et al. Parallel altitudinal clines reveal trends in adaptive evolution of genome size in *Zea mays*. PLOS Genet. 2018; 14: e1007162.

58. Rosato M, Chiavarino A, Naranjo C, Hernandez J, Poggio L. Genome size and numerical polymorphism for the B chromosome in races of maize (*Zea mays ssp. mays*, Poaceae). Am J Bot. 1998; 85: 168.

59. Lia V V, Confalonieri VA, Poggio L. B chromosome polymorphism in maize landraces: adaptive vs. demographic hypothesis of clinal variation. Genetics. 2007; 177: 895–904. https://doi.org/10.1534/genetics.107.075440 PMID: 17954923

60. Freitas FO, Bendel G, Allaby RG, Brown TA. DNA from primitive maize landraces and archaeological remains: implications for the domestication of maize and its expansion into South America. J Archaeol Sci. 2003; 30: 901–908.

61. Kistler L, Yoshi Maezumi S, De Souza JG, Przelomska NAS, Costa FM, Smith O, et al. Multiproxy evidence highlights a complex evolutionary legacy of maize in South America. Science. 2018; 362: 1309–1313. https://doi.org/10.1126/science.aav0207 PMID: 30545889

62. López MG, Fass M, Rivas JG, Carbonell-Caballero J, Vera P, Puebla A, et al. Plastome genomics in South American maize landraces: chloroplast lineages parallel the geographic structuring of nuclear gene pools. Ann Bot. 2021; 128:115–125.

63. Perales HR, Brush SB, Qualset CO. Landraces of maize in Central Mexico: an altitudinal transect. Econ Bot. 2003; 57: 7–20.